

$$Q1) \quad \{\sigma_i, \sigma_j\} = 2\delta_{ij} \sigma_0$$

$$[\sigma_i, \sigma_j] = 2i \epsilon_{ijk} \sigma_k.$$

$$\text{Prove that } \sigma_i \sigma_j = \delta_{ij} \sigma_0 + i \epsilon_{ijk} \sigma_k.$$

$$(1) \quad \sigma_i \sigma_j + \sigma_j \sigma_i = 2\delta_{ij} \sigma_0$$

$$(2) \quad \sigma_i \sigma_j - \sigma_j \sigma_i = 2i \epsilon_{ijk} \sigma_k$$

$$(1) \text{ implies } \sigma_j \sigma_i = 2\delta_{ij} \sigma_0 - \sigma_i \sigma_j$$

$$(2) \text{ implies}$$

$$\sigma_i \sigma_j - [2\delta_{ij} \sigma_0 - \sigma_i \sigma_j] = 2i \epsilon_{ijk} \sigma_k$$

$$2\sigma_i \sigma_j = 2\delta_{ij} \sigma_0 + 2i \epsilon_{ijk} \sigma_k$$

$$\text{Hence } \sigma_i \sigma_j = \delta_{ij} \sigma_0 + i \epsilon_{ijk} \sigma_k$$

as desired.

Q2) Prove $(\vec{A} \cdot \vec{\sigma})(\vec{B} \cdot \vec{\sigma}) = (\vec{A} \cdot \vec{B})\sigma_0 + i(\vec{A} \times \vec{B}) \cdot \vec{\sigma}$.

A, B commute with σ 's.

Label $\vec{A} = A_i \hat{e}_i$

similar w/ σ 's and B .

We have $A_i \sigma_i B_j \sigma_j = A_k B_k \sigma_0 + i \epsilon_{lmn} A_l B_n \sigma_m$

$$+ i A_l B_n \underbrace{\epsilon_{lmn} \sigma_m}_{\frac{1}{i}(\sigma_l \sigma_n - \delta_{ln} \sigma_0)}$$

$$= A_k B_k \sigma_0 + A_l B_n (\sigma_l \sigma_n - \delta_{ln} \sigma_0)$$

$$= A_k B_k \sigma_0 + A_l \sigma_l B_n \sigma_n - A_l B_l \sigma_0$$

$$= A_l \sigma_l B_n \sigma_n$$

as desired.

Q3) Two spin system in state $|+-\rangle$ s.t

$$S_z |+-\rangle = \frac{\hbar}{2} |+-\rangle$$

$$S_{2z} |+-\rangle = -\frac{\hbar}{2} |+-\rangle.$$

Rotate $|+-\rangle$ state by $\frac{\pi}{2}$ about $\hat{x}$ axis, express resulting state in terms of basis kets $|+\rangle, |-\rangle, |+\rangle, |-\rangle$.

$$\begin{aligned} {}^{(1)}R\left(\frac{\pi}{2}, \hat{x}\right) &= e^{-i\frac{\pi}{2}\hbar \cdot \frac{1}{2}\hat{x} \cdot \sigma} \\ &= e^{-i\frac{\pi}{4}\sigma_x} \\ &= \left(\cos\frac{\pi}{4}\sigma_0 - i\sin\frac{\pi}{4}\sigma_x\right) \\ &= \begin{pmatrix} \cos\frac{\pi}{4} & -i\sin\frac{\pi}{4} \\ -i\sin\frac{\pi}{4} & \cos\frac{\pi}{4} \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \\ -\frac{i}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}. \end{aligned}$$

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

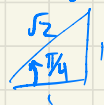
$$\sigma_y = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$$

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$\frac{\pi}{4}$ ccw

$$\cos\frac{\pi}{4} = \cos(45^\circ) = \frac{1}{\sqrt{2}}$$

$$\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$$



$$\text{Hence } {}^{(1)}R_1\left(\frac{\pi}{2}, \hat{x}\right) \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{1}{\sqrt{2}} (|+\rangle - i|-\rangle)$$

$${}^{(1)}R\left(\frac{\pi}{2}, \hat{x}\right) \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} -i \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} (-i|+\rangle + |-\rangle)$$

Express in terms of $|+\rangle, |-\rangle$ kets.

Then, taking tensor product

$$\begin{aligned} R_1 |+\rangle_1 \otimes R_2 |-\rangle_2 &= \frac{1}{2} (|+\rangle_1 - i|-\rangle_1) \otimes (-i|+\rangle_2 + |-\rangle_2) \\ &= \frac{1}{2} (-i|++\rangle + |+-\rangle - |-+\rangle - i|--\rangle). \end{aligned}$$

As desired. Since $\vec{S}_1, \vec{S}_2$ commute.

Q4) 2) Show that singlet state

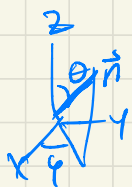
$$|\psi\rangle = \frac{1}{\sqrt{2}} (|+-\rangle - |-+\rangle)$$

can be written as

$$\frac{1}{\sqrt{2}} (|\vec{n}+\rangle_1 \otimes |\vec{n}-\rangle_2 - |\vec{n}-\rangle_1 \otimes |\vec{n}+\rangle_2)$$

for any direction $\vec{n}$. $\vec{n}^2 = 1$.

$$\vec{n} \equiv (\cos\theta\sin\phi, \sin\theta\sin\phi, \cos\theta)$$



for an arbitrary axis.

Any arbitrary rotation can be written as

$$R(\theta, \phi) = \begin{pmatrix} \cos\frac{\theta}{2} e^{-i\phi/2} & -\sin\frac{\theta}{2} e^{-i\phi/2} \\ \sin\frac{\theta}{2} e^{i\phi/2} & \cos\frac{\theta}{2} e^{i\phi/2} \end{pmatrix}$$

$\uparrow$
 $|\vec{n}, +\rangle$

$\uparrow$
 $|\vec{n}, -\rangle$

$$\text{So } |\psi\rangle = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} \cos\frac{\theta}{2} e^{-i\phi/2} \\ \sin\frac{\theta}{2} e^{i\phi/2} \end{pmatrix}_1 \otimes \begin{pmatrix} \sin\frac{\alpha}{2} e^{-i\beta/2} \\ \cos\frac{\alpha}{2} e^{i\beta/2} \end{pmatrix}_2 - \begin{pmatrix} \sin\frac{\theta}{2} e^{-i\phi/2} \\ \cos\frac{\theta}{2} e^{i\phi/2} \end{pmatrix}_1 \otimes \begin{pmatrix} \cos\frac{\alpha}{2} e^{-i\beta/2} \\ \sin\frac{\alpha}{2} e^{i\beta/2} \end{pmatrix}_2 \right]$$

When $\theta=0, \varphi=0$ / $\alpha=0, \beta=0$ (Both along z-dir)

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix}_1 \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix}_2 - \begin{pmatrix} 0 \\ 1 \end{pmatrix}_1 \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix}_2 \right]$$

Now $|\vec{n}+\rangle_1 \otimes |\vec{n}-\rangle_2$ means $\theta=\alpha, \varphi=\beta$,

$$= \left[\cos\frac{\theta}{2} e^{-i\varphi/2} |+\rangle_1 + \sin\frac{\theta}{2} e^{i\varphi/2} |-\rangle_1 \right] \\ \otimes \left[\sin\frac{\theta}{2} e^{-i\varphi/2} |+\rangle_2 + \cos\frac{\theta}{2} e^{i\varphi/2} |-\rangle_2 \right]$$

$$= \cos\frac{\theta}{2} \sin\frac{\theta}{2} e^{-i\varphi} |+-\rangle + \cos^2\frac{\theta}{2} |-+\rangle + \sin^2\frac{\theta}{2} |-+\rangle + \sin\frac{\theta}{2} \cos\frac{\theta}{2} e^{i\varphi} |--\rangle$$

Similarly $|\vec{n}-\rangle_1 \otimes |\vec{n}+\rangle_2$ yields

$$= \sin\frac{\theta}{2} \cos\frac{\theta}{2} e^{-i\varphi} |++\rangle + \cos^2\frac{\theta}{2} |-+\rangle + \sin^2\frac{\theta}{2} |+-\rangle + \cos\frac{\theta}{2} \sin\frac{\theta}{2} e^{i\varphi} |--\rangle$$

When summed (subtracted)

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left[\cos^2\frac{\theta}{2} |+-\rangle + \sin^2\frac{\theta}{2} |-+\rangle - \cos^2\frac{\theta}{2} |-+\rangle - \sin^2\frac{\theta}{2} |+-\rangle \right]$$

$$= \frac{1}{\sqrt{2}} \left[(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}) |+-\rangle - (\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}) |-+\rangle \right]$$

$$= \frac{(\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2})}{\sqrt{2}} [|+-\rangle - |-+\rangle]$$

$$\text{but } \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} = \cos(2\cdot\frac{\theta}{2}) = \cos\theta.$$

Normalizes to 1.

b) Use above to show that

$$e^{-i\mu/\hbar \vec{n} \cdot \vec{S}} |\psi\rangle = |\psi\rangle$$

($|\psi\rangle$ is rotation invariant) by an angle μ , $\vec{S}_1 + \vec{S}_2 = \vec{S}$ and $\vec{S}_1, \vec{S}_2$ commute.

Just show for both $|n+\rangle$ and $|n-\rangle$,

$$\left(\cos \frac{\mu}{2} \sigma_{n1} - i \sin \frac{\mu}{2} \vec{n} \cdot \vec{\sigma}_{n1} \right) |\vec{n}+\rangle_{n1} = |\vec{n}+\rangle_{n1} \quad \text{for } n=1/2$$

corresponding to
Hilbert space 1/2

sim for $|\vec{n}-\rangle$. $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$$\begin{aligned} \vec{n} \cdot \vec{\sigma} &= \cos \psi \sin \theta \sigma_x + \sin \psi \sin \theta \sigma_y + \cos \theta \sigma_z \\ &= \begin{pmatrix} \cos \theta & \cos \psi \sin \theta - i \sin \psi \sin \theta \\ \cos \psi \sin \theta + i \sin \psi \sin \theta & -\cos \theta \end{pmatrix} \end{aligned}$$

Rot. op is then $e^{i\frac{\mu}{2} \vec{n} \cdot \vec{\sigma}_{n1}} e^{i\frac{\mu}{2} \vec{n} \cdot \vec{\sigma}_{n2}}$ is $\vec{S} = \vec{S}_1 + \vec{S}_2$

$$\left(\cos \frac{\mu}{2} \sigma_0 - i \sin \frac{\mu}{2} \vec{n} \cdot [\vec{\sigma}_{n1} + \vec{\sigma}_{n2}] \right)$$

$$\begin{aligned} R|+-\rangle &= R|n+\rangle_1 \otimes |n-\rangle_2 \\ &= R \begin{pmatrix} \cos \frac{\theta}{2} e^{i\psi/2} \\ \sin \frac{\theta}{2} e^{i\psi/2} \end{pmatrix}_1 \otimes \begin{pmatrix} \sin \frac{\theta}{2} e^{-i\psi/2} \\ -\cos \frac{\theta}{2} e^{i\psi/2} \end{pmatrix}_2 \end{aligned}$$

R acts independently on both states

$$\left(\cos \frac{\mu}{2} \sigma_0 - i \sin \frac{\mu}{2} \vec{n} \cdot \vec{\sigma}_{(1)} \right) \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\varphi/2} \\ \sin \frac{\theta}{2} e^{i\varphi/2} \end{pmatrix}_1$$

$$= \begin{pmatrix} \cos \frac{\mu}{2} - i \sin \frac{\mu}{2} \cos \theta & -i \sin \frac{\mu}{2} (\cos \varphi \sin \theta - i \sin \varphi \sin \theta) \\ -i \sin \frac{\mu}{2} (\cos \varphi \sin \theta + i \sin \varphi \sin \theta) & \cos \frac{\mu}{2} + i \sin \frac{\mu}{2} \cos \theta \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\varphi/2} \\ \sin \frac{\theta}{2} e^{i\varphi/2} \end{pmatrix}$$

$\swarrow \sin \theta e^{i\varphi}$ $\nearrow \sin \theta e^{-i\varphi}$

$$= \begin{pmatrix} \cos \frac{\theta}{2} (\cos \frac{\mu}{2} - i \sin \frac{\mu}{2} \cos \theta) e^{-i\varphi/2} + \sin \frac{\theta}{2} (-i \sin \frac{\mu}{2} \sin \theta) e^{-i\varphi} e^{i\varphi/2} \\ \cos \frac{\theta}{2} (-i \sin \frac{\mu}{2} \sin \theta) e^{i\varphi} e^{-i\varphi/2} + \sin \frac{\theta}{2} (\cos \frac{\mu}{2} + i \sin \frac{\mu}{2} \cos \theta) e^{i\varphi/2} \end{pmatrix}$$

$\underbrace{e^{i\varphi/2}}_{e^{i\varphi/2}}$

$$= \begin{pmatrix} \cos \frac{\theta}{2} \cos \frac{\mu}{2} e^{-i\varphi/2} - i \cos \frac{\theta}{2} \sin \frac{\mu}{2} \cos \theta e^{-i\varphi/2} - i \sin \frac{\theta}{2} \sin \frac{\mu}{2} \sin \theta e^{-i\varphi/2} \\ -i \sin \frac{\mu}{2} \cos \frac{\theta}{2} \sin \theta e^{i\varphi/2} + \sin \frac{\theta}{2} \cos \frac{\mu}{2} e^{i\varphi/2} + i \sin \frac{\theta}{2} \cos \theta \sin \frac{\mu}{2} e^{i\varphi/2} \end{pmatrix}$$

$\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}$ $2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$ $2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$

$$= \begin{pmatrix} \cos \frac{\theta}{2} \cos \frac{\mu}{2} e^{-i\varphi/2} - i \cos \frac{\theta}{2} \sin \frac{\mu}{2} e^{-i\varphi/2} + i \cos \frac{\theta}{2} \sin \frac{\mu}{2} \cos^2 \frac{\theta}{2} e^{-i\varphi/2} - \frac{1}{2} i \sin^2 \frac{\theta}{2} \sin \frac{\mu}{2} \cos^2 \frac{\theta}{2} e^{i\varphi/2} \\ - \frac{1}{2} i \sin^2 \frac{\theta}{2} \cos^2 \frac{\theta}{2} \sin \frac{\mu}{2} e^{i\varphi/2} + \sin \frac{\theta}{2} \cos \frac{\mu}{2} e^{i\varphi/2} + i \sin \frac{\theta}{2} \sin \frac{\mu}{2} \cos^2 \frac{\theta}{2} e^{i\varphi/2} - i \sin^3 \frac{\theta}{2} \sin \frac{\mu}{2} e^{i\varphi/2} \end{pmatrix}$$

$$= \begin{pmatrix} \cos \frac{\theta}{2} \cos \frac{\mu}{2} e^{-i\varphi/2} - i \cos \frac{\theta}{2} \sin \frac{\mu}{2} \left(\cancel{\cos \frac{2\theta}{2}} + \sin \frac{2\theta}{2} \right) e^{-i\varphi/2} \\ \sin \frac{\theta}{2} \cos \frac{\mu}{2} e^{i\varphi/2} - i \sin \frac{\theta}{2} \sin \frac{\mu}{2} \left(\cancel{\cos \frac{2\theta}{2}} + \sin \frac{2\theta}{2} \right) e^{i\varphi/2} \end{pmatrix}$$

$$= \begin{pmatrix} \cos \frac{\theta}{2} e^{-i\frac{\mu}{2}} e^{-i\varphi/2} \\ \sin \frac{\theta}{2} e^{-i\frac{\mu}{2}} e^{i\varphi/2} \end{pmatrix}$$

$R_{(i)} |\vec{n}+\rangle_{(i)}$ transformation
 $= e^{-i\frac{\mu}{2}} |\vec{n}+\rangle_{(i)}$

Similarly: $R_{(i)} |\vec{n}-\rangle_{(i)}$

$$= \begin{pmatrix} \cos \frac{\mu}{2} - i \sin \frac{\mu}{2} \cos \theta & -i \sin \frac{\mu}{2} \sin \theta e^{-i\varphi} \\ -i \sin \frac{\mu}{2} \sin \theta e^{i\varphi} & \cos \frac{\mu}{2} + i \sin \frac{\mu}{2} \cos \theta \end{pmatrix} \begin{pmatrix} -\sin \frac{\theta}{2} e^{-i\varphi/2} \\ \cos \frac{\theta}{2} e^{i\varphi/2} \end{pmatrix}$$

$$= \begin{pmatrix} -\sin \frac{\theta}{2} \cos \frac{\mu}{2} e^{-i\varphi/2} + i \sin \frac{\mu}{2} \sin \frac{\theta}{2} (\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}) e^{-i\varphi/2} - i \sin \frac{\mu}{2} \cdot 2 \sin \frac{\theta}{2} \cos^2 \frac{\theta}{2} e^{-i\varphi/2} \\ i \sin \frac{\mu}{2} \cdot 2 \sin^2 \frac{\theta}{2} \cos \frac{\theta}{2} e^{i\varphi/2} + \cos \frac{\mu}{2} \cos \frac{\theta}{2} e^{i\varphi/2} + i \sin \frac{\mu}{2} \cos \frac{\theta}{2} (\cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2}) e^{i\varphi/2} \end{pmatrix}$$

$$= \begin{pmatrix} -\sin \frac{\theta}{2} \cos \frac{\mu}{2} e^{-i\varphi/2} + i \sin \frac{\mu}{2} \sin \frac{\theta}{2} \cancel{\cos^2 \frac{\theta}{2}} e^{-i\varphi/2} - i \sin \frac{\mu}{2} \sin^3 \frac{\theta}{2} e^{-i\varphi/2} - \cancel{2i} \sin \frac{\mu}{2} \cdot \sin \frac{\theta}{2} \cos^2 \frac{\theta}{2} e^{-i\varphi/2} \\ \cancel{2i} \sin \frac{\mu}{2} \cos \frac{\theta}{2} \sin^2 \frac{\theta}{2} e^{i\varphi/2} + \cos \frac{\mu}{2} \cos \frac{\theta}{2} e^{i\varphi/2} + i \sin \frac{\mu}{2} \cos^3 \frac{\theta}{2} e^{i\varphi/2} - \cancel{i \sin \frac{\mu}{2} \cos \frac{\theta}{2} \sin^2 \frac{\theta}{2} e^{i\varphi/2}} \end{pmatrix}$$

$i \sin \frac{\mu}{2} \cos \frac{\theta}{2} e^{i\varphi/2}$

$$= \begin{pmatrix} -\sin\frac{\theta}{2} e^{-i\varphi/2} e^{i\mu/2} \\ \cos\frac{\theta}{2} e^{i\varphi/2} e^{i\mu/2} \end{pmatrix} = e^{i\mu/2} |\vec{n}-\rangle_{(u)}$$

Thus $R_1 |\vec{n}+\rangle, R_2 |\vec{n}-\rangle_2 = e^{i\mu/2} e^{-i\mu/2} |\vec{n}+\rangle_1 |\vec{n}-\rangle_2$

$$= |\vec{n}+\rangle_1 |\vec{n}-\rangle_2$$

and same w $|\vec{n}-\rangle_1 |\vec{n}+\rangle_2$, so $|\psi\rangle$

is rotation invariant.

Q5) Magnetic induction field $\vec{B}$
direction def'd by θ, φ .

At time $t=0$ e^- exists w spin $\uparrow$ $\vec{B}$.

If e^- state defined by

$$|\psi\rangle = A|+\rangle + B|-\rangle$$

Find A, B ($|+\rangle \equiv$ spin up / $|-\rangle \equiv$ spin down)

Recall that $\hat{H} = -\frac{q}{m} \vec{B} \cdot \vec{S}$

in SE:
$$i\hbar \begin{pmatrix} \dot{\psi}_+ \\ \dot{\psi}_- \end{pmatrix} = -\frac{q}{m} \frac{\hbar}{2} \vec{B} \cdot \vec{\sigma} \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}$$

$\vec{B}$ -dir'n def'd by θ, φ . $\vec{B} = B(\cos\varphi\sin\theta, \sin\varphi\sin\theta, \cos\theta)$

Spin along $\vec{B} \Rightarrow R|\psi\rangle R(\theta) \underbrace{\begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{|+\rangle} = \begin{pmatrix} \cos\frac{\theta}{2} e^{i\varphi/2} \\ -\sin\frac{\theta}{2} e^{-i\varphi/2} \end{pmatrix}$

Thus $|\theta, \varphi\rangle = \begin{pmatrix} \cos\frac{\theta}{2} e^{i\varphi/2} \\ -\sin\frac{\theta}{2} e^{-i\varphi/2} \end{pmatrix}$

can we project onto the basis

$$|\psi\rangle = A|+\rangle + B|-\rangle$$

$$= |+\rangle \langle + | \theta, \varphi \rangle + |-\rangle \langle - | \theta, \varphi \rangle$$

$$| \psi(t=0) \rangle = \cos \frac{\theta}{2} e^{i\varphi/2} |+\rangle - \sin \frac{\theta}{2} e^{-i\varphi/2} |-\rangle$$

@ time $t=0$.

Q6 - Skip, learn for next
this is straightforward...

Q7) $t=0$ e^- in state $|+\rangle$.

$$\vec{B} = B \hat{x}. \quad |\vec{B}| = 100 \text{ gauss.}$$

How many seconds for spin to flip?

$$\hat{H} = -\frac{q}{m} \vec{B} \cdot \vec{s} = -\frac{q\hbar}{2m} B \sigma_x$$

$$i\hbar \begin{pmatrix} \dot{\psi}_+ \\ \dot{\psi}_- \end{pmatrix} = -\frac{q\hbar B}{2m} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix}$$

$$\therefore \dot{\psi}_+ = \frac{iqB}{2m} \psi_-$$

$$\text{Let } \omega = \frac{qB}{2m}.$$

$$\dot{\psi}_- = \frac{iqB}{2m} \psi_+.$$

$$\begin{aligned} \ddot{\psi}_+ &= i\omega \cdot i\omega \psi_+ \\ &= -\omega^2 \psi_+ \end{aligned}$$

$$\psi_+(t) = A e^{i\omega t} + B e^{-i\omega t}, \text{ then}$$

$$\begin{aligned} \psi_-(t) &= A i\omega t e^{i\omega t} - B i\omega t e^{-i\omega t} \\ &= A t e^{i\omega t} - B t e^{-i\omega t}. \end{aligned}$$

$$|\psi\rangle = [A e^{i\omega t} + B e^{-i\omega t}] |+\rangle + [A t e^{i\omega t} - B t e^{-i\omega t}] |-\rangle.$$

$$\text{At } t=0 \quad |\psi(t=0)\rangle = |+\rangle, \text{ so } A+B=1, \quad A-B=0.$$

$$A=B=\frac{1}{2}.$$

$$e^{i\omega t} + e^{-i\omega t} = 2\cos\omega t$$

$$e^{i\omega t} - e^{-i\omega t} = 2i\sin\omega t$$

cancels $\frac{1}{2}$ factors.

$$|\Psi\rangle = \cos\omega t|+\rangle + i\sin\omega t|-\rangle$$

(normalized).

Spin flips when $\omega t = 0 \rightarrow \omega t = \frac{\pi}{2}$,

$$t = \frac{\pi}{2\omega}$$

$$= \frac{\frac{\pi}{4B}}{2 \frac{\mu}{2\mu}} = \frac{\pi\mu}{4B}$$

onto addition Hw

Q1) - 456 Part Quiz

$$\hat{H} = A \hat{\vec{j}}_1 \cdot \hat{\vec{j}}_2$$

$$\vec{j}^2 = (\vec{j}_1 + \vec{j}_2)^2 = j_1^2 + j_2^2 + 2\vec{j}_1 \cdot \vec{j}_2$$

$$\hookrightarrow \hat{H} = \frac{A}{2} (\vec{j}^2 - \vec{j}_1^2 - \vec{j}_2^2)$$

1) A > 0 Quantum numbers of G_{SS} :

$$\text{Recall } J^2 |j, m\rangle = \hbar^2 j(j+1)$$

$$\text{Thus } \hat{H} |j_1, m_1; j_2, m_2\rangle = \frac{\hbar^2 A}{2} [(j_1 + j_2)(j_1 + j_2 + 1) - j_1(j_1 + 1) - j_2(j_2 + 1)]$$

For $A > 0$, there is a minimum when $j_1 = j_2 = 0$ which creates a unique, lowest-energy ground state.

2) A < 0: For $A < 0$, quantum numbers are the same except that the ground state is degenerate, there are multiple values of j that can give same G_{SS} unless $J = j_{\max}$, then it is unique.

If S_1, S_2 have same spin state (j_{max}), then

$$\hat{U}|j_{\text{max}}\rangle = \frac{A_b}{2} (2j(j+1) - 2j(j+1))$$

$$j_{\text{tot}} = 2j, 2j-1, 2j-2, \dots, 1, 0.$$